

# Worksheet — Air Resistance Simulation

Simulation 15 · use with *Simulation - Air resistance.html* · A.1 Kinematics: qualitative effect of air resistance on a projectile (drag  $F_d = \frac{1}{2}\rho C_d A v^2$ ); goes with throwing / filming a ping-pong ball or shuttlecock beside a tennis ball

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

## How to use this worksheet

Open **Simulation - Air resistance.html** in a browser (it works offline). Choose an **object**, the **speed** and the **angle**, then  Launch. The **solid** path has air resistance; the **dashed** path is the same launch in a vacuum.

**Part 1** before the lab ( $\approx 15$  min) · **Part 2** during the lab ( $\approx 15$  min) · **Part 3** homework ( $\approx 30$  min).

Drag always points **opposite to the velocity** and grows with  $v^2$ . At SL you must describe its effects; the numbers help you see them.

## Part 1 — Pre-lab: predict, then test

Tennis ball,  $25 \text{ m s}^{-1}$  at  $45^\circ$ . Predict **before** you launch, then test and explain.

| Question                                                                          | My prediction (before) | What the simulation showed | Physics explanation |
|-----------------------------------------------------------------------------------|------------------------|----------------------------|---------------------|
| 1. How does the real path differ from the vacuum path? Sketch both.               |                        |                            |                     |
| 2. On the way up, is the acceleration smaller than, equal to or larger than $g$ ? |                        |                            |                     |
| 3. Which takes longer: going up or coming down?                                   |                        |                            |                     |
| 4. In which direction does the drag act at the top of the path?                   |                        |                            |                     |

5. Calculate the range in a vacuum for  $25 \text{ m s}^{-1}$  at  $45^\circ$ . By what percentage is the real range shorter?

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## Part 2 — During the lab: light and heavy balls

**Class.** Throw a ping-pong ball (or shuttlecock) and a tennis ball at about the same speed and angle; film them if you can.

**Simulation:**  $10 \text{ m s}^{-1}$  at  $45^\circ$  for both.

|                                          | ping-pong ball | tennis ball | vacuum |
|------------------------------------------|----------------|-------------|--------|
| range $R$ / m                            |                |             |        |
| max height $H$ / m                       |                |             |        |
| landing angle (steep / similar)          |                |             |        |
| terminal speed $v_t$ / $\text{m s}^{-1}$ |                |             | —      |

**1. Describe the shape of the ping-pong ball's path. How is it different from the tennis ball's?**

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**2. Explain why the ping-pong ball is affected much more. Use  $F_d/m$ .**

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**3. Sketch the path of a shuttlecock from the class (or video) and compare with the simulation.**

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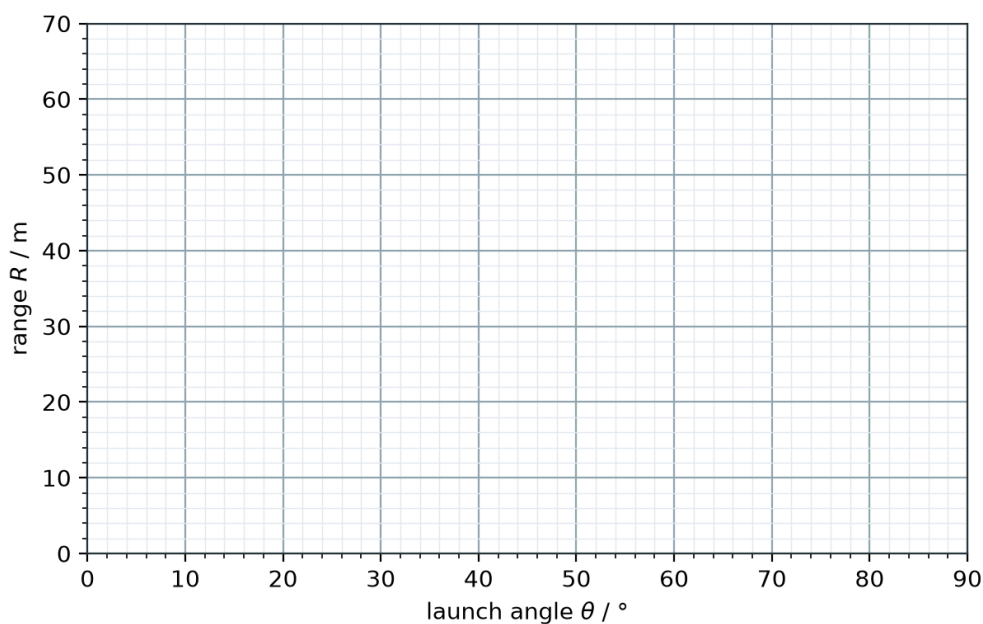
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## Part 3 — Homework: analyse

Tennis ball,  $25 \text{ m s}^{-1}$ . Record each angle and copy both ranges.

| $\theta / ^\circ$ | 20 | 30 | 35 | 40 | 45 | 50 | 60 |
|-------------------|----|----|----|----|----|----|----|
| R with air / m    |    |    |    |    |    |    |    |
| R vacuum / m      |    |    |    |    |    |    |    |

1. Plot both ranges against  $\theta$  on the same axes.



2. At what angle is the range with air largest? Why is it less than  $45^\circ$ ?

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3. Use  $mg = \frac{1}{2}\rho C_d A v_t^2$  to calculate the terminal speed of a ping-pong ball ( $m = 2.7 \text{ g}$ ,  $d = 40 \text{ mm}$ ,  $C_d = 0.47$ ,  $\rho = 1.20 \text{ kg m}^{-3}$ ).

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4. Calculate the drag force on a tennis ball ( $m = 58 \text{ g}$ ,  $d = 67 \text{ mm}$ ,  $C_d = 0.55$ ) at  $25 \text{ m s}^{-1}$  and compare it with its weight.

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**5. For which object is the vacuum model best: ping-pong ball, shuttlecock, tennis ball or steel ball?**

**Explain.**

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